Kelvin Schouten

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Profile

MSc Quantitative Finance student with a BSc in Mathematics and six months of independent industrial modelling research. Experience in numerical modelling, Monte Carlo uncertainty analysis, and validation against experimental measurements.

Education

MSc Quantitative Finance 2026-Present
Tilburg University, Netherlands
Relevant coursework: Empirical Finance, Panel Data Econometrics, Valuation & Risk

BSc Mathematics 2026
Utrecht University, Netherlands
BSc research (2025): (Non-)periodicity in fractional order differential systems | Supervisor: Paul Zegeling

HBO Applied Mathematics (not completed) 2021
Higher professional education; continued with a BSc Mathematics

Applied Research Experience

Research and Development: Two-Phase Flow Modelling & Uncertainty Quantification 2026 - Six-month project
Independent project for an industrial company | Food-industry pipelines

- Reviewed recent scientific literature, implemented and adapted published two-phase-flow models for the company's pipeline equipment and operating conditions.
- Modelled system requirements and cleaning performance across different flow regimes and industrial operating scenarios.
- Implemented Monte Carlo simulations to quantify how uncertainty in model inputs affects predictions of system requirements and cleaning outcomes.
- Validated model predictions against experimental measurements to assess predictive performance under industrial operating conditions.
- Calibrated the model to improve its accuracy and reliability in predicting the emptying and cleaning of the pipelines.

Quantitative Finance Project

Empirical Finance Data Analysis MSc coursework
Python | Fama–French factors and industry portfolio returns

- Processed and analysed Fama-French factor data and industry portfolio returns in Python for regression-based empirical finance exercises.

Technical Skills

Programming & tools: Python, Jupyter notebooks, Git, GitHub

Quantitative methods: Numerical modelling, Monte Carlo simulation, parameter estimation, uncertainty propagation, model validation

Statistics & finance: Regression analysis, econometrics, probability theory, stochastic processes, financial return analysis

Research: Scientific literature review, mathematical model adaptation, technical reporting

Additional Experience

Research Assistant 2020-2019
The Hague University of Applied Sciences

- Analysed student performance data to investigate differences in academic outcomes; documented and presented findings.
- Credited contributor to the 2019 research report *Ondersteuningsnetwerken* for the Inclusive Education research group.

Mathematics & Physics Tutor 2017-2025
Altena College and independent tutoring

- Developed tailored exercises and provided mathematics and physics instruction to secondary school students.

Languages

Dutch: Native | **English:** Proficient